C-command effects in binding third-person reflexives and pronoun in Turkish

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This study investigates the role that c-command plays in long-distance binding of anaphors and pronouns in Turkish. Turkish is a head-final, agglutinating SOV language spoken by approximately 75 million speakers worldwide. In Turkish pronouns and anaphors are preverbal, which arguably provides a better testing ground to investigate how structural information is used to resolve binding dependencies [1]. To lay the groundwork for online studies, we report the results of an offline antecedent choice task with two third-person reflexives (bare reflexive *kendi,* possessive-marked *kendisi*) and one third-person pronoun (*o*) [2,3]. Previous work on Turkish reflexives largely focused on the locality properties of *kendi* and *kendisi* [4-6]. It has been argued that *kendisi* does not have to be c-commanded by its antecedent [3], but to our knowledge, no study has directly investigated the role of c-command on binding of Turkish reflexives. Our results corroborate previous claims that neither *kendi* nor *kendisi* is strictly governed by Principle A [4-6]. We also show that nonetheless, long-distance binding of the two reflexive forms is sensitive to c-command, but they do not show uniform sensitivity to it. The bare reflexive *kendi* is more sensitive to c-command than the possessive-marked *kendisi*. Furthermore, we present the first experimental evidence that the third-person pronoun *o* in Turkish can refer to a non-local or a sentence-external antecedent, as predicted by Principle B. Experiments looking at the real-time processing of these pronouns are underway.

In an offline **antecedent choice** task ($N_{subjects}$=37, N_{items}=36), we investigated Turkish speakers' choice of antecedents for the two reflexives and the simple pronoun *o.* We used a 2x3 within-subject design crossing the type of PRONOUN (*kendi, kendisi* or *o*) and C-COMMAND (whether the non-local antecedent c-commanded the pronoun, or not). In the C-COMMAND conditions, the non-local antecedent was the subject of the matrix clause and the pronoun was embedded inside a non-finite nominalized clause (1a). In the NO C-COMMAND conditions, the non-local antecedent was the possessor in a genitive-possessive structure (1b). Participants had three options: they could select the local, non-local, and a sentence-external antecedent ('other'). Participants could choose any options that were possible.

Both the bare reflexive *kendi* and the possessive-marked reflexive *kendisi* preferred local antecedents over non-local antecedents, but *kendi* had an increased likelihood for local antecedents than *kendisi* (*p*'s < .01) (see Table 1). For non-local antecedents, both of the reflexives and the pronoun preferred c-commanding to non-c-commanding antecedents (*p*'s ≤ .01). Within reflexives, *kendi* showed a much larger effect of c-command than did *kendisi* in binding non-local antecedents (*p*'s ≤ .01). The pronoun *o* was generally not taken to refer to the local antecedent. Surprisingly, *kendisi* was less likely to refer to the local antecedent in the c-commanding configurations than in the non-c-commanding configurations (*p* < .05).

These data show that c-command plays a role in non-local binding of Turkish reflexives. *Kendi* and *kendisi* were both less likely to take non-local antecedents in non-c-commanding configurations, and this tendency was greater for *kendi* than for *kendisi*. This offline pattern mirrors results in Mandarin [7]: in both Mandarin and Turkish, the bare reflexive form shows greater sensitivity to c-command than does the feature-marked form. Data collection for online studies investigating the time-course of these effects is still ongoing.

References:

[1] Kush, D., & Phillips, C. (2014). *Frontiers in Psychology, 5*. **[2]** Göksel, A. & Kerslake, C. (2005). New York: Routledge. **[3]** Kornfilt, J. (2001). In P. Cole, G. Hermon & C.-T. James Huang (eds.). **[4]** Gračanin-Yüksek, M. et al. (2017). *Journal of Psycholinguistic Research, 46*(6). **[5]** Özturhan, M. (2016). Unpublished MA thesis. **[6]** Özbek, A., & Kahraman, B**. (**2016). *Dil ve Edebiyat Dergisi, 13*(1). **[7]** Dillon, B. et al. (2016). *Frontiers in Psychology, 6.*

Materials: Colored words show the pronoun manipulation.

1. a. C-Command:

Müdür-Ø sekreter-in … o-nu/kendi-ni/kendi-si-ni öv-düğ-ün-ü
principal-NOM secretary-GEN … them-/self-/self-POSS-ACC praise-FN-3SG-ACC
bil-iyor-Ø
know-PROG-3SG
'The principal knows that the secretary is praising them/themself/themself all the time.'

b. No C-Command:

Müdür-ün sekreter-i … o-nu/kendi-ni/kendi-si-ni öv-üyor-Ø
principal-GEN secretary-POSS … them-/self-/self-POSS-ACC praise-PROG-3SG
'The principal's secretary is praising them/themself/themself all the time.'

Table 1. The percentages (%) for the choice of local, non-local and sentence-external ('other') antecedents in the offline antecedent choice task.

	kendi		***kendisi***		***o***	
	C-C	**NoC-C**	**C-C**	**NoC-C**	**C-C**	**NoC-C**
Local	90.10	93.75	79.17	91.15	15.10	16.15
Non-local	72.40	19.27	83.33	52.60	87.50	78.13
Other	5.21	10.42	14.58	20.31	81.25	84.38